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COE431 – Project I I

HydroGuard
(Smart Water Quality Bottle)

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PROJECT DECLARATIONS

1. It is declared that this project is original and is resulted as a group effort of the project members only, unless otherwise specified and in the document and references. Moreover, this work in its entirety is never submitted in any form to quality for another degree or diploma at any university/institution and I/We undertake not to submit the same in future for qualifying and obtaining any other degree from this or any other university/institution.

2. I/We affirm to have applied ethics to the design process and in the selection of the project design. I/We also affirm to have considered the safety of the public to be paramount and have addressed this in the design wherever applicable.

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LIST OF DESIGN CONSIDERATIONS

Please fill in the following table as per the design considerations of the project.

The impact of the project on:	Aspects	Applicability		Extent of Applicability				
Economy	The project covers feasibility of development, production and operational cost.	Yes	No	1	2	3	4	5
Environment	The project has an effect on environment (negative to positive)	Yes	No	1	2	3	4	5
Ethic	The project does not have any obvious potential conflicts with a professional Code of Ethics arising from the development or use of the product	Yes	No	1	2	3	4	5
Health and Safety	Safety and health of public is being considered while designing and developing the project and it does not violate any existing laws of health and safety	Yes	No	1	2	3	4	5
Society	The project is beneficial for the community and society as a whole. The impact on the society is considered	Yes	No	1	2	3	4	5

LIST OF STANDARDS FOLLOWED IN THE PROJECT

Please provide a list of standards followed in the design and implementation of the project

S. No.	Standard Name	Standard Body
1	**IEEE 802.11™**: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications for Wi-Fi communication	IEEE
2	**IEEE 802.15.1™**: Bluetooth wireless communication standard (BR/EDR and BLE)	IEEE
3	**ISO 5667**: Water quality — Sampling methods	ISO
4	**ISO/IEC 25010**: Systems and Software Quality Requirements and Evaluation (SQuaRE)	ISO/IEC
5	**ISO/IEC 27001**: Information security management for cloud data handling	ISO/IEC
6	**WHO Guidelines for Drinking-water Quality**: Thresholds for contaminants and water safety classification	World Health Organization (WHO)
7	**I²C and UART Communication Protocol Standards**: For sensor–microcontroller interfacing	NXP / Industry Standard
8	**RoHS Directive (2011/65/EU)**: Restriction of Hazardous Substances in electrical and electronic equipment	European Union (EU)
9	**SI Units**: Standard units for pH, NTU, mg/L, mS/cm, °C in measurement reporting	Bureau International des Poids et Mesures (BIPM)

CERTIFICATE OF APPROVAL

This is to certify that the project titled "_____" accomplished by_____, Reg. ID. _____, under the supervision of Dr._____, Ajman University, UAE, is fully acceptable, in quality and scope, as ProjectX as required by the study plan of the degree of *B.Sc. in Computer Engineering*.

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ABSTRACT

Clean and safe drinking water remains a global challenge, and many existing portable testers fail to provide accurate or meaningful interpretation of water conditions. This project presents **HydroGuard**, a portable IoT-based water quality monitoring device designed to offer **real-time sensing and AI-assisted evaluation**. The final prototype integrates **four sensors**—pH, turbidity, temperature, and total dissolved solids (TDS)—controlled by an ESP32 microcontroller within a compact, custom-designed housing.

In Project II, the system was upgraded to include a **local web dashboard** hosted on the ESP32, where users can read live sensor data without the need for external applications. An **AI evaluation layer**, implemented through the OpenAI API, analyzes the collected parameters and generates a clear safety assessment based on WHO guidelines, along with suggested actions for improving water quality. The device operates on a rechargeable battery, includes an OLED screen for immediate display, and is optimized for field use through stable hardware integration and improved enclosure design.

The enhanced HydroGuard system demonstrates a reliable, low-cost, and user-friendly solution for household inspection, outdoor testing, academic use, and emergency water quality assessment.

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CHAPTER 1

INTRODUCTION

Chapter 1: INTRODUCTION TO PROJECT

1.1 Overview

Clean drinking water is a basic human right, but millions of individuals are struggling to ensure that their water is safe daily in this world. It is under such urgency that this project proposes the Smart Water Quality Testing Bottle, which is innovative, compact, portable, and aims to test the safety/quality of drinking water in a real-time manner.

The system combines several water quality sensors along with a display on the board, AI algorithms and wireless communications into a slick bottle-type design. It is constructed to target people and communities that need a fast, precise, and convenient technique to determine water safety wherever they are. This report summarizes the development, application, and capabilities of this solution, which brings together embedded systems, artificial intelligence, and mobile communication to provide a potent and useful instrument of water sensing [19].

1.2 Statement of Problem

The problem of water pollution is still of concern all over the world and health problems are the most common conditions for people to get Ebola diarrhea, cholera etc. caused by water pollution [19]. Although laboratory checks are accurate, they are expensive, time consuming and not accessible to use on a daily or on-the-go basis. Moreover, most households, travelers and campers do not have equipment and knowledge to test water quality prior to their consumption.

Conventional handheld test kits consist of one to two parameters, can only be interpreted manually and do not provide digital reports. The issue is that there is no compact multi-sensor system available that would be able to:

- Multiparameter water analysis Test
- Show or broadcast the results on-demand
- Increase accuracy and interpretation with the help of AI
- Be cheap, portable and newbie-friendly

This project will be able to fill that gap by creating an intelligent water testing bottle that will be able to use several sensors and intelligent algorithms to recognize tainted water and alert the user in addition to capturing the data to be saved and retrieved either later or remotely.

1.3 Project Specifications

1.3.1 Hardware Specifications

- ESP32 microcontroller with built-in Wi-Fi
- pH sensor (analog input)
- TDS sensor (analog input)
- Turbidity sensor (analog input)
- DS18B20 digital temperature sensor
- 0.96" OLED display (I2C interface)
- Rechargeable battery with TP4056 charging module
- Internal metal bottle enclosure with waterproof sensor placement
- On-board web server hosted locally on the ESP32

1.3.2 Software Specifications

- Arduino IDE firmware written in C/C++
- Real-time data acquisition from all sensors
- Calibration offsets for pH and TDS
- Dynamic web dashboard showing live readings (HTML + Chart.js)
- Local API endpoint “/data” for JSON sensor output
- AI analysis using OpenAI API to generate water-safety verdict
- OLED visual status indicator (“SAFE” / “NOT SAFE”)

1.3.3 System Functional Specifications

- Measure pH, TDS, turbidity, and temperature in real time
- Display readings on OLED and on the web interface
- Store latest reading in structured data format
- Send collected readings to OpenAI every 2 minutes
- Receive summarized safety analysis and recommended actions
- Operate fully on battery power during field testing

1.4 Motivation/Justification

The primary drive in having such a project is to fill in the gap of the available water testing devices that are not readily open to the public and that they can utilize. Concerned, people require an easy and fast mechanism to know when the water is safe to ensure that diseases are not encountered in the countryside, on the road or even in cases of post-disaster assistance.

Existing water test machines are either:

- Laboratory and costly, or
- Manual and non-automatic Incomplete

This project democratizes water safety testing, by combination of cheap microcontrollers, smart sensors, and edge AI. It is especially helpful at:

- Inhabitants of areas that have unmet water supplies
- Red cross operations Here, the red cross is supposed to provide emergency relief teams to assist them.
- Out bounders, Hikers and Campers
- Environ mental education

1.5 Relevance to Sustainable Development Goals (SDGs)

This project directly supports the **United Nations Sustainable Development Goals (SDGs)**, particularly:

- **SDG 6: Clean Water and Sanitation**

  This project facilitates the vision of ensuring availability and sustainable management of water to all by creating a smart, portable, and cost-effective solution that monitors the quality of water. Real time capacity of the system to detect contaminants in the water supply like pH out of balance, high turbidity, unsafe TDS level, and unusual conductivity helps ensure drinking water quality. This can enhance good health, sanitation, and responsible water management among the developed and developing countries [19].

- **SDG 3: Good Health and Well-Being**

  Access to clean water is directly associated with elimination of water borne diseases like cholera, diarrhea and typhoid. The project guarantees reduced health risks by affording direct water safety classification and alert so that the

world goal of healthy lives and promotion of well-being could be achieved globally [19].

- **SDG 9: Industry, Innovation, and Infrastructure**

  This system achieves innovation in environmental monitoring with merging of AI, IoT and multi-sensors in a small form factor. It promotes the implementation of intelligent technologies in infrastructure construction and sustainability of the environment [22].

By aligning with these goals, the Smart Water Quality Testing Bottle not only serves as a technological innovation but also as a tool for global sustainable development and public health improvement.

1.6 Project Applications

There are lots of applications for the Smart Water Quality Testing Bottle:

- Home Use: To determine the quality of tap or well water by the family.
- Outdoor: Hikers, campers or outdoors travelers to wild locations.
- Disaster Response: The areas that have been affected by floods or earthquake and water pollution are very prevalent.
- Undeveloped Regions: It can be utilized by the NGOs to confirm water sources in underserved or rural regions.
- Smart Cities: Incorporation into IoT systems of urban water quality testing.
- Research & Education: To study environment and create awareness through school/ college programs.

1.7 Discussion on Standards and Constraints

Engineering Standards Overview:

- **ISO 5667**: Standards for water quality sampling.
- **ISO/IEC 25010**: For software/system quality (reliability, usability, etc.)
- **IEEE 802.11:** Incorporated for enabling wireless communication [8].
- **ISO/IEC 27001**: If cloud storage is used, for securing user data.
- **I2C / UART Protocols**: For sensor communication.
- **WHO Guidelines for Drinking Water Quality**: Used as reference thresholds [20].

Design Constraints Overview:

It is through some practical and technical limitations that the design of the Smart Water Quality Testing Bottle has been determined, serving to strategically enforce its effectiveness, usability and cost. Given that the system is to be operated as portable apparatus using batteries, power conservation is of utmost importance so as to have long operations before admitting the need to recharge the batteries. Calibration of the sensors is also essential because the precision of the measurements should be observed over a wide range of environmental settings and through long-term operation. Form factor and size also present a limitation, since all hardware must be packaged into a small, bottle-shaped package that does not reduce performance. It must also be affordable; the price of its production must be kept low so that the device can be affordable to the highest number of users, especially in the developing populations [21]. Also, it must be waterproof to avoid water splashes or other related damages to fragile electronics in the field under different conditions, and at the same time, to survive such accidents. Lastly, as the system uses AI-based analysis, the target level of accuracy can be only reached after training the model on a variety and sufficiently representative data set so that the safety classification can ensured in any situation in practice.

1.8 Organization of the Report

There is the first chapter that serves as a basis to the Smart Water Quality Testing Bottle by giving an overview of the project. This paper starts by giving the significance of the issue of water quality on human health and safety and then proceeds to the gist of the problem that this project will address. The chapter describes the specifications of the system, motivation of the project and identifies the applications of the system and beneficiaries in the real world. It also talks about the major engineering standards to be considered when designing the product, physical and technological constraints, that influenced work on the system.

The second chapter immerses into the literature review, with the emphasis on the state of the water quality monitoring technologies. It includes the following associated technologies: multi-parameter sensors, Internet of Things (IoT) based devices, AI in environmental sensing, and small embedded systems. Review of existing projects, commercial products and scholarly work is also given in the chapter and conclusions

based on those were analyzed and research gaps that the current project attempts to bridge were established. This part forms the technical background and theoretical support to the design decisions that will be provided in the forthcoming sections of the report.

The following third chapter shifts the focus on problem analysis and dissects the core problems with real-time water testing. It enumerates the criteria which can be used to choose sensors, power management, environmental resilience, and the problems of data interpretation. The chapter contains an analysis of the components and tests the possibility of integration of a bottle-sized form. It also covers the technical and logistical choices made according to the issues identified during the first planning.

In the fourth chapter, the methodology and design of the system is given. It presents the entire block diagram of hardware architecture, then a flow diagram of the required steps execution to complete the system functions. This chapter will be split into two major sections of its design, hardware system design (microcontroller, sensors, display, battery, enclosure) and software system design (sensor data acquisition, AI model integration, user interface logic). It also describes the data gathering, training, testing and implementation of the AI component that would determine whether water is safe or unsafe with the input related to the sensor.

The fifth chapter dwells upon the development of tools and techniques. It describes the tools, libraries and platforms that will be used to program the hardware, to train machine learning models and to simulate. As an example, it can use Arduino IDE to develop firmware, TensorFlow Lite-based machine learning models can be run on an embedded system, and cloud platforms with optional web dashboards are used. It also addresses methodologies in testing water parameters, data logging as well as testing of sensor accuracy.

Lastly, a report that is a detailed conclusion is given in the sixth chapter. It goes back to the original goals, outlines the major project outputs, and mentions the performance of the system on what the intended goals were. The chapter also summarizes the restrictions which are present now, including calibration slippage, power execution, or sensor life, and provides an idea of future improvement. This can be better AI accuracy,

waterproofing design, the support of additional water parameters, or similar connection with public health systems.

CHAPTER 2

LITERATURE REVIEW

Chapter 2: LITERATURE REVIEW

This chapter is an overview of available technologies, projects and available solutions regarding water quality monitoring. It contains two main parts: (1) Related Technologies, mentioning the enabling technologies supporting the creation of intelligent water quality checkers, and (2) Related Projects that review past work and commercial products in the area. The chapter ends with a view of shortcomings that have pervaded the current solutions and which this project tries to fill.

2.1 Related Technologies

In this section of the report, we will discuss the various technologies that are related to **HydroGuard**.

2.1.1 Multi-Parameter Water Quality Sensors

The current water quality testing relies on a variety of sensors that determine physical, chemical, biological parameters of water. The common environmental sensors include pH sensor, TDS (Total Dissolved Solids) sensor, turbidity sensor, temperature sensor, conductivity sensor, and the dissolved oxygen sensor.

Li et al. (2022) provides the information that these sensors may be integrated with microcontroller-based systems with their analog or digital data interfaces (I2C, UART) which makes it possible to obtain and process the data in real-time. These sensors must be precise and longstanding, which are critical in creating water safety assessments that people can rely on. Developments in MEMs technology now allow these to be small and cheap, they can thus be fitted into portable systems such as the proposed smart bottle [14], [23].

2.1.2 Artificial Intelligence in Water Analysis

Artificial intelligence is a game changer when it comes to reading sensor data. The currents of the raw water measure which can be quite inconsistent, and noisy rendering manual interpretation incomplete. Such researchers as Kumar et al. (2021) have suggested the application of machine learning (ML) models, including decision trees, support vector machines (SVM), and artificial neural networks (ANN), to assort water quality on a scale of classes, including but not limited to safe or contaminated.

In addition, publicly available environmental databases or sets of labeled data by the government health agencies can be used as training ML models. The resulting trained model can be subsequently deployed to edge with TinyML frameworks such as TensorFlow Lite that enables inference on microcontrollers without cloud connection [13], [22].

2.1.3 Internet of Things (IoT) in Environmental Monitoring

IoT technologies have transformed the process of monitoring and sharing the environmental parameters. Commonly, in the smart water monitoring the sensors communicate with microcontrollers (e.g., ESP32 or Arduino) which relays information using Wi-Fi, Bluetooth, or LoRaWAN technologies to cloud-based services. Zhang et al. (2023) also states that IoT allows building distributed sensor networks that may send feedback to its users in real-time and officials.

The mobile apps and dashboards could be used privately to visualize the trend, memorize a warning, and keep the past data. In such a project, IoT integration produces such effects as the data concerning water quality could be visualized on a smartphone or the browser [19], [21].

2.1.4 Embedded Systems for Portable Testing

Small smart devices are at their core embodied in embedded systems. Microcontrollers like the ESP32 have in-built wireless modules, multiple analog inputs, and allow running of AI inference models. When combined with small displays (e.g. OLED or TFT) and battery packs they facilitate the creation of “field ready” tools that are self-powered.

Sensor compatibility, power efficiency and hardware flexibility of microcontrollers indirectly influence the structure of portable testing kits. The development is simplified with the use of open-source platforms such as the Arduino ecosystem and edge AI frameworks that allow local data processing and classification [6].

2.1.5 Cloud and Mobile Integration

Many contemporary environmental monitoring systems have integration or support apps to increase their availability and accessibility. The users can see real-time readings, trend graphs, see some historical changes and be alerted in case of unsafe readings. The two most popular frameworks to use cloud dashboards and IoT is Blynk, Thingspeak or Firebase. Custom web or mobile application interfaces are Flask and React Native.

In the given project, the connectivity feature will allow users to get AI-enhanced interpretations of water safety data remotely, which is in line with the idea of modern and comfortable user experience [2], [19].

2.2 Related Projects

2.2.1 IoT-Based Water Quality Monitoring System (2020)

Patel et al. presented a system that is based on the application of IoT to present a water monitoring system with the use of Arduino and a variety of sensors (pH, TDS, turbidity). Data was transmitted to ThingSpeak to monitor it in real time [15], [21]. Despite being applicable in fixed installations (such as water tanks), it did not have AI capabilities, portability, and the capability to view the display on device.

2.2.2 Smart Water Bottle for Athletes (2021)

In this student-led project at MIT, a smart bottle was involved, which monitors hydration through sensors of water consumption. Although a creative idea in fitness, it was not dealing with real contamination of water or even testing of quality. It demonstrates that electronic bottle electronics is possible and usable [26].

2.2.3 AI-Powered Water Safety Classifier (2022)

In Germany, researchers trained a neural network in water tests and installed it on a Raspberry Pi to label the water to be safe or not. The system was however bulky, needed mains power and lacked any form factor that would allow it to be easily portable, or perform real-time tasks to a non-technical user [25].

2.2.4 Commercial Portable Test Kits (e.g., TestDrop Pro)

Various consumer products are marketed that make purported single-drop (usually through the electrical conductivity) tests of water quality. Nevertheless, they test mostly TDS, which cannot be considered a full-fledged level of water security. Few of them offer the AI interpretation and live feedback [17].

Table 1: Comparison of Related Works in Water Quality Monitoring

Project / Product	Sensors Used	Approx. Cost	Accuracy	Wireless Support	Portability
IoT-Based Water Quality Monitoring System (Patel et al., 2020)	pH, TDS, Turbidity	Low (~$50)	Medium (±5–10%)	Wi-Fi (ThingSpeak)	Low – Fixed installation
Smart Water Bottle for Athletes (MIT, 2021)	Flow / Hydration sensors	Medium (~$80)	High for hydration measurement	Bluetooth	High – Portable
AI-Powered Water Safety Classifier (Müller et al., 2022)	pH, TDS, Turbidity	High (~$150)	High (≥90%)	None	Low – Bulky, mains-powered
TestDrop Pro (Commercial)	Electrical Conductivity (EC)	Low (~$50)	Low – Only EC measurement	None	High – Handheld
Proposed System (Smart Water Quality Testing Bottle)	pH, TDS, Turbidity, Temp, Conductivity, DO	Low–Medium (~$80–100)	High (>90% with AI)	Wi-Fi / Bluetooth	High – Bottle-shaped, portable

2.3 Limitations/Problems in Literature Work

Though several research studies and commercial activities have been performed in water quality monitoring, upon critical analysis most of the currently existing system have left behind the complete range of the real-world applicability in terms of usability, intelligence as well as portability. Such constraints are as follows:

Narrow parameter coverage: most systems are based on one sensor (e.g. TDS or pH) and are insufficient to generate a full picture of water safety. This may result to erroneous findings and dangerous inferences.

Absence of AI Integration: Majority of the devices show raw convictions of the sensors, and they have no intelligent interpretation or safety classifications. This slows down using it by non-technical users and carries a low level of accuracy in making decisions.

Portability: Many of these academic prototypes are too big, power-consuming, or even dependent upon additional hardware to be practical everyday tools or in the field.

Weak Real-Time Feedback: The systems normally take cloud processing or manual analysis that increases response time and does not work under offline conditions.

Poor User Interface: In numerous designs, there are insufficient feedback (such as a display or a safety indicator), which makes them look garbled or not suitable to standard users.

Internet Dependency: The cloud reliant systems cannot be used in the rural areas or remote regions where testing of water is required the most.

Calibration and reliability concerns: Low-cost systems can be subject to sensor drift, and variation in calibration, but such issues have not been discussed in the literature.

Therefore, the proposed project consists of a multi-sensor bottle with compact size, AI functioning, offline operation, real-time safety classification, and the entire testing user-friendly experience in real life water testing.

CHAPTER 3

PROBLEM ANALYSIS

Chapter 3: PROBLEM ANALYSIS

3.1 Analysis of the Problem Specification

Clean and safe drinking water remains a major global challenge; more than two billion people consume water that does not meet WHO safety standards. Contaminated water may contain harmful microorganisms, chemical pollutants, and suspended particles that can lead to serious health issues such as cholera, typhoid, and diarrhea. Although laboratory-based water testing technologies are accurate, they are slow, require specialized equipment and trained personnel, and are not suitable for quick decision-making in homes, outdoor environments, or emergency situations.

Portable water-testing kits offer partial improvement in accessibility, but these devices usually measure only one or two parameters (such as pH or TDS) and often present raw values without interpretation. Such outputs require technical knowledge to understand, making them unsuitable for non-expert users and prone to misinterpretation. Additionally, most low-cost devices do not integrate multiple sensor readings into a single safety decision, nor do they provide real-time digital visualization or intelligent analysis.

In this project, the Smart Water Quality Testing Bottle addresses these gaps by combining multiple essential sensors—pH, TDS, turbidity, and temperature—inside a compact bottle-mounted design. The ESP32 microcontroller processes sensor readings, hosts a local web dashboard, and communicates with an AI model through an online API to generate an easy-to-understand safety verdict. The OLED screen provides immediate classification (“SAFE” / “NOT SAFE”), while the built-in Wi-Fi allows users to view detailed graphs and AI explanations through a local webpage without requiring any external mobile application.

To achieve these goals, the system must satisfy both functional and non-functional requirements. Functionally, it must perform accurate multi-parameter sensing, process data in real time, call an AI model for interpretation, and show results on both the OLED

display and web interface. Non-functional requirements include portability, water-resistant sensor placement, reliable power supply, stable Wi-Fi hosting, and the ability to operate even when cloud services are temporarily unavailable. Tools used in development include the Arduino IDE for firmware, sensor libraries for calibration, web-dashboard libraries, and basic CAD tools for designing the bottle enclosure. Together, these features aim to create a practical, user-friendly, and field-ready solution for real-time water quality evaluation.

3.2 Required Components and Calibration

The Smart Water Quality Testing Bottle is also composed of both some software and hardware features which jointly come together so as to measure, process and present accurate results of the water quality.

Software wise, firmware development is possible through the Arduino IDE and PlatformIO allowing a multi-sensor and multi-input programming platform on the ESP32 microcontroller. This artificial intelligence model is developed in Python with TensorFlow/Keras and converted in to TensorFlow Lite for Microcontrollers to make the model run effectively on the ESP32 with minimal memory requirements. Optional connections are provided by means of using either Firebase or Blynk services, connected logs, notifications, and historical monitoring can be reified by use of an internet connection. With Figma, user interface designs of the integrated OLED display and the mobile application are designed in a manner that information is well presented and friendly.

On the software front, the system core is an ESP32-WROOM microcontroller, which is a dual-core, and 240 MHz is capable of supporting integrated Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 [5]. The module was selected because it supports multiple sensors and supports processing AI inference and wireless connectivity, all without having to add additional modules. The sensing array contains:

- pH Sensor (SEN0161-V2) A sensor that measures acidity/alkalinity of the material provided based on a range of 0-14 pH and accuracy of measuring pH is +- 0.1 pH [3].
- Turbidity Sensor (SEN0189) -Measures clarity of water between 0 and 1000 NTU and has ± 5 accuracy [4].

- Total Dissolved Solids (TDS) Sensor - Detects dissolved solids with 0-1000 ppm patterns and an accuracy of +/-10 per cent.
- Electrical Conductivity Sensor (DFR0300) Ionic content 0 to 20 mS/cm with an accuracy of ±2%.
- Temperature Sensor (DS18B20) 12-bit; temperature range of between -55 to 125 C and an accuracy of +/? 0.5 degrees C.
- Dissolved Oxygen (DO) Sensor- Monitors the amount of oxygen present with a range between 0-20 mg/L and a net accuracy of 0.2 mg/L [14].
- Optional Heavy Metal Sensor Electrochemical sensor to detect metal, lead or copper.

The sensors connect to ESP32 in a combination of analog connections (pH, turbidity, TDS, conductivity, DO, heavy metals), I2C to the OLED display, and 1-Wire to the temperature sensor. It has a 0.96″ OLED screen (128×64 resolution) to display both the numerical values and AI safety classifications ahead of time on the device. The power is provided with 2000 3000 mAh rechargeable Lithium-ion battery with a charge management module to ensure safety.

Calibration will make the device produce correct and identical measurements. Standard buffer solutions of pH 10.0, 7.0, and 4 and used in calibrating the pH sensor. Turbidity sensor is compared with Formazin reference standards at various NTU values. TDS and conductivity signals are calibrated by means of KCl of known concentration. The process of calibration of the DO sensor involves the use of water that has oxygen in 100% and sodium sulfite in 0. The temperature sensor is calibrated before the thermometers found in the laboratory. The calibration coefficients are saved in a non-volatile memory in order to maintain the device sensitivity to the passage of time after which measures are given to recalibrate in case drift is observed [18], [23].

3.3 Decision Making based on Problem Analysis

The ergonomics of the Smart Water Quality Testing Bottle was driven by the requirements to eliminating the particular drawbacks of the other technologies on the water testing market as well as maintaining the system small, efficient, and simple to carry. The choice of Espressif ESP32-WROOM microcontroller was a tactical decision

since it has the computing power to process data across multiple sensors at once, has wireless connectivity available to use to implement optional remote monitoring capability, and can run an on-device embedded AI model locally without having to resort to cloud processing. This would guarantee the ability of the device to perform well both on connected and offline operations, which is effective when the setting may be in a remote locality, disaster response areas, or even outdoor trips that can not have internet around them.

The selection of a multi-sensor method as opposed to using a single parameter of measurement was important to increase the accuracy and reliability of the findings. Single parameters like pH, turbidity, TDS, conductivity, temperature, dissolved oxygen, and heavy metals will give only partial answers concerning the quality of water but when they are analyzed together in the process of AI analysis, results are always more reliable. Such an integrated solution removes the issue of false confidence which may arise when single or few water quality indicators are monitored.

Embedded AI/ML model supports the possibility to process the readings of all sensors in real-time and classify water as fit or unfit according to the WHO criteria. This removes the reliance upon the user to make his/her own interpretation of raw measurements, thus removing the likelihood of error, particularly amongst non expert operators. The choice of offline AI operation instead of cloud computing processing of information makes the system operable in any location without concern of data privacy, and optional online capabilities including data logging and analysis of trends over time is possible when connected.

Mechanically, the enclosure should be made small, enclosed, portable, and enable steady sensor installation to get a measured reading. It is not bound to a specific container shape since design can be adapted easily to variations in deployment scenarios and the design does not lock a project to a particular design by forcing the project to adopt the specific structure. Wiring and locations of sensors are carefully considered internally in order to avoid electrical interferences and cross-contamination of measurements. A lithium-ion battery with ability to be recharged was selected because it gives longer periods of operation and use of low-power modes of firmware consumption when the device is idle will assist in saving energy.

The risk analysis was used to determine any possible vulnerability. Time drift in sensors is compensated by the periodicity of calibration prompts and the calibration data is saved in non-volatile memory to allow the effectiveness of accuracy over the period of time between uses. The risk of the battery becoming empty is decreased with power saving modes and a battery icon level in the display. The risk of AI misclassification is mitigated through a training set with a diverse large dataset and having a confidence score in the results that will trigger a retest should the confidence be on the lower end. The environmental protection measures against destruction can be computerized through splash resistance sealing, protective gasket, and conformality coating electronic appliances.

Table 2: Comparison between Traditional and Modern Monitoring

Feature	Traditional Portable Kits	Smart Water Quality Testing Bottle
Parameters Measured	Typically 1–2 (e.g., pH, TDS)	6–7 including pH, turbidity, TDS, conductivity, temperature, DO, heavy metals
Result Interpretation	Manual, requires expertise	Automatic AI-based classification
Operation Mode	Fully offline	Offline with optional online data logging
Real-Time Alerts	Not available	Yes, on-device safe/unsafe indication
Calibration Tracking	Manual, user-dependent	Automatic reminders and stored calibration data
Data Logging	Rarely available	On-device storage and optional cloud sync
Usability	Risk of error for non-experts	Designed for both experts and general users

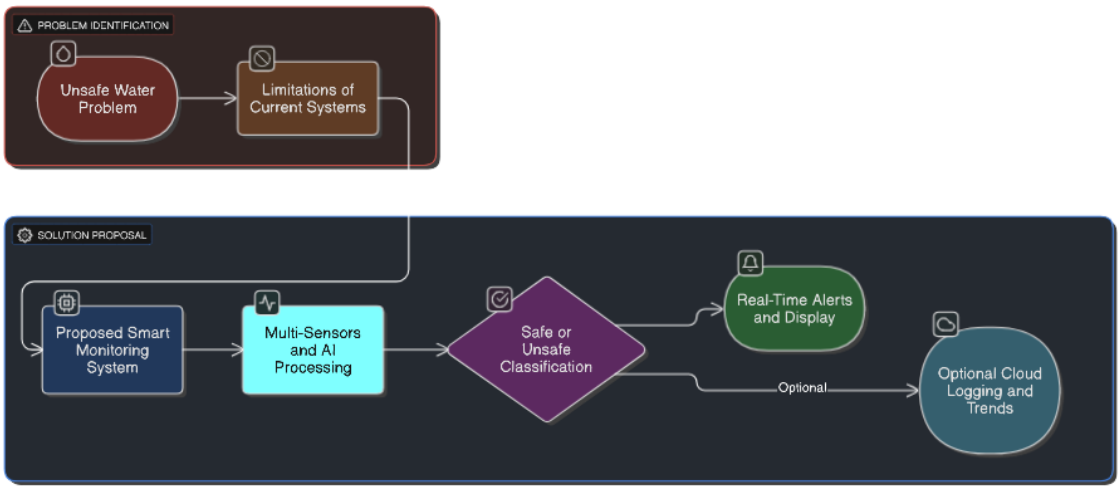


Figure 1 - Problem identification and proposed solution workflow for the Smart Water Quality Testing Bottle.

CHAPTER 4
PROJECT DESIGN AND METHODOLOGY

Chapter 4: DESIGN AND METHODOLOGY

4.1 Overall Project Design

The Smart Water Quality Testing Bottle is developed as a built-in system to consist of several calibrated sensors, an ESP32 microcontroller, and embedded AI process to enable a real-time evaluation of the drinking water safety. The design is portable, offline capable, and user-friendly, hence can be used in different contexts including monitoring at home, undertaking activities outdoors, and making field checks in remote locations.

The centering point of the system is the ESP32-WROOM microcontroller to receive six to seven sensors reading pH, turbidity, total dissolved solids (TDS), electrical conductivity, temperature, dissolved oxygen (DO) and, if necessary, heavy metals. Their readings are processed before they are read to eliminate any temperature effect, eliminate any noise and normalize the data. The ESP32 is directly running an AI/ML model-trained on a dataset of water samples, with TensorFlow Lite for Microcontrollers providing real-time water classification locally as being either Safe or Unsafe by WHO standards [20].

The outcomes appear in real time on an OLED 0.96-inch screen and the results will feature a visual or audible alert, should it prove unsafe. Non-obligatory Bluetooth or Wi-Fi can be designed to access the device to sync the results with a mobile application in order to track history, provide graphical analysis, and reminders that the calibration is due. A rechargeable lithium-ion battery powers it, and an onboard charging module controls it to be used in the field longer.

4.1.1 Block Diagram

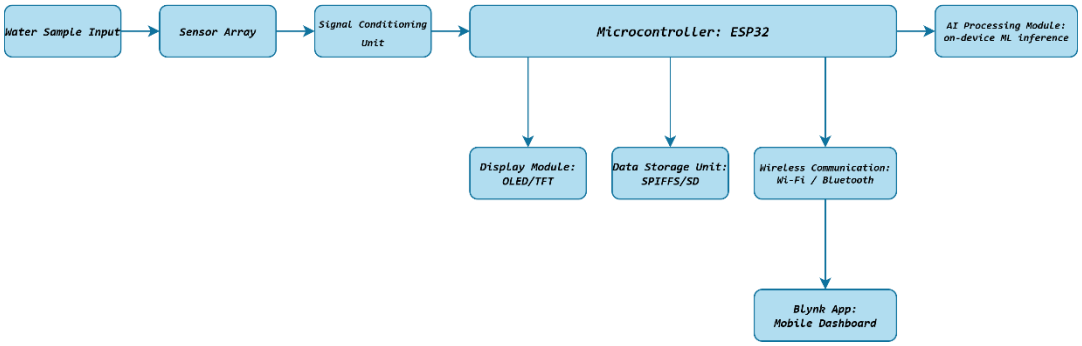


Figure 2 - Block diagram describing the functionality of the system.

Figure 4.1.1 is the block diagram of the overall functionality of the Smart Water Quality Bottle. The core part of the system is the ESP32 microcontroller which organizes all the actions. It starts by the user gathering a sample of water in the supplied inbuilt container. When the sample has been identified, the ESP32 turns on the sensor array that is comprised of pH, turbidity, TDS, conductivity, temperature, dissolved oxygen, and heavy metals sensors. Raw sensor data is recorded, sent to the preprocessing module, and here noise is filtered, values are normalized and temperature compensation performed. The smoother data is then inputted to the AI-based classification feature that decides on whether the sample is in the set safety walls or not. The outcomes of the classification will be illustrated on the OLED screen and sent through Bluetooth or Wi-Fi to the corresponding cell phone application of the user. Meanwhile, readings are kept locally and can be uploaded (optionally) to a cloud database where they are tracked on an ongoing basis. Peripheral modules, including charging, power management circuits and optional calibration guidance interface are also depicted in the block diagram to provide reliable, energy efficient operation.

4.1.2 Flow Diagram

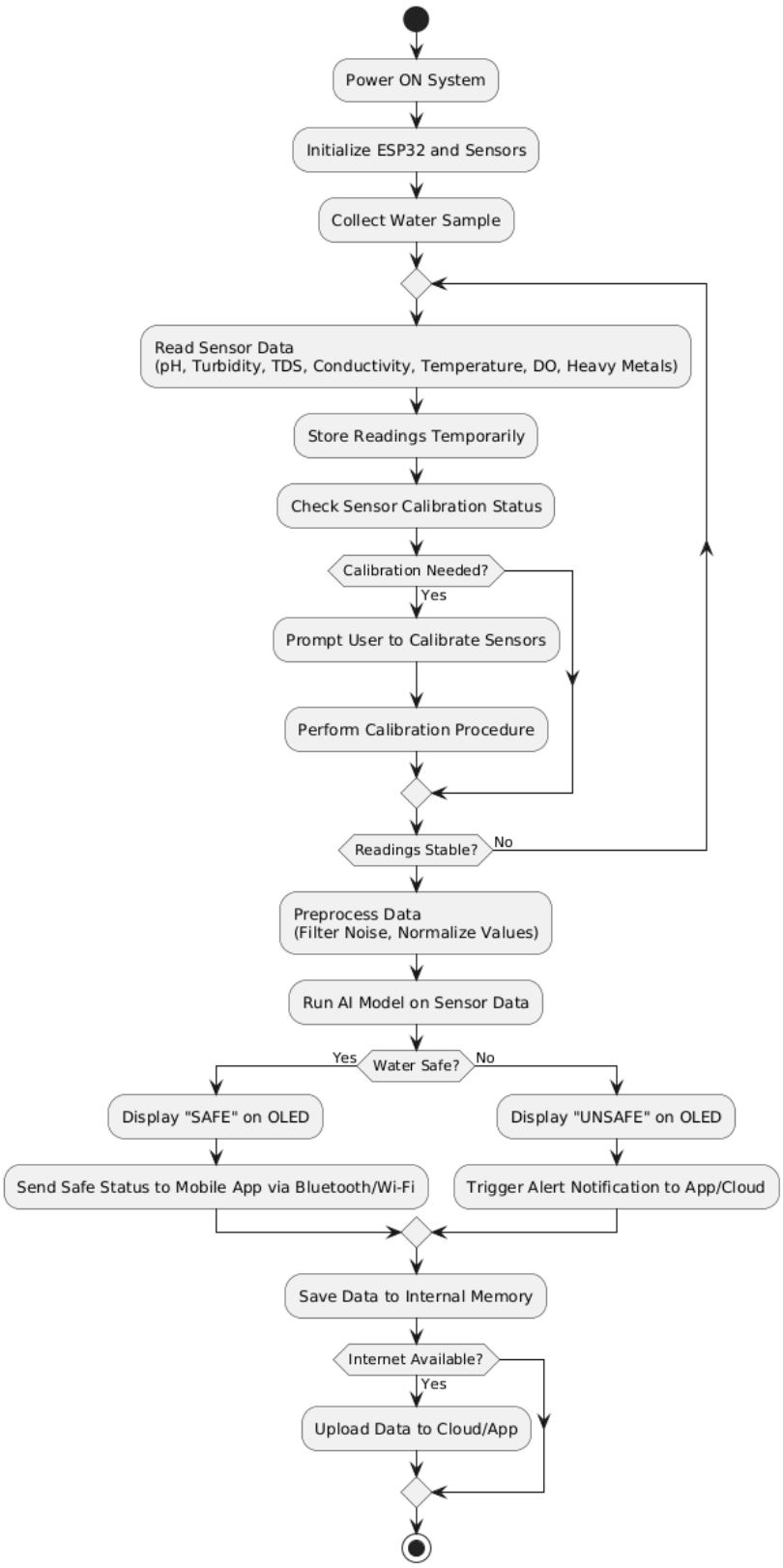


Figure 3 - Flow Diagram

As the flowchart drawn in Figure 4.1.2 shows, the process cycle of the Smart Water Quality Bottle represents the chronological order of its work, starting with initialization and ending with the delivery of final results. When powered up, all hardware on the ESP32 is initialized, this includes the sensor array, display module and wireless communication interface. The user is then asked to sample and store a water sample by the system. The sensor readings are taken serially with a stability check done to maintain consistency of the measurements. In case of calibration, the user is taken through the procedure until he reaches stable readings. The preprocessing is done to eliminate noise, exclude outliers, normalize data and correct the readings to the impacts of the temperature. The cleaned data is subjected to analysis, being fed to the onboard AI model, which decides which category of water to use: Safe or Unsafe. The outcome is displayed on the OLED screen, and unsafe findings cause instant warnings to the related mobile application. The readings and the classification of all readings are stored on non volatile memory then sign in to the cloud when and where there is internet and monitoring done remotely. The device functions and operates correctly, efficiently, and in real-time because of this structured flow; thus investors get results of credible water safety.

4.2 Methodology

This section describes the complete implementation process of the Smart Water Quality Bottle, from sample collection to budget planning. The methodology ensures accurate water quality analysis, user-friendly operation, and cost-effective deployment.

4.2.1 Sample Collection

It starts with the obtaining of a water sample in the built-in container. The design of bottle removes the chances of contamination during sampling as the bottle is designed with a sealed cap and pre-washed inside. The sampling can either be triggered manually through a push button or pre set intervals automatically as dictated by the mode of operation.

4.2.2 Sensor Data Acquisition

When the samples are detected, ESP32 microcontroller turns on the built-in sensor array that contains:

- PH Sensor- acidity/alkalinity [3].
- Turbidity Sensor – detection of water clarity [4].
- TDS Sensor - detection of total dissolved solids [14].
- Conductivity Sensor-sensor to with ionic concentration [14].
- Temperature Sensor - to read and to compensate on temperature [14].
- Dissolved Oxygen Sensor: – saturation of the oxygen [14].
- Heavy Metal Sensor-detection of lead and mercury and other contaminants [14].

Resulting values of the sensors are read serially by ESP32 [5] and stored in volatile memory where they undergo preprocessing.

4.2.3 Calibration Process

Calibration of the sensors is necessary within an interval. The system checks stored calibration data at the startup. In case the calibration is necessary, the OLED display and the mobile app take the customer through regular procedures with the help of known solutions of calibration [18]. They also include the calibration parameters which are saved in EEPROM to survive reboots.

4.2.4 Data Preprocessing

Data are collected and then passed through some steps of data processing prior to analysis:

- Noise Filtering: Stabilising readings through moving averages.
- Outlier Removal:Disregarding outlying values outside acceptable limits.
- Normalization: reducing the values to the expected range of the AI model.
- Temperature Compensation: The pH and conductivity are adjusted according to measured temperature [13], [22].

4.2.5 AI-Based Classification

The AI model, which is embedded on the ESP32 or in the cloud, conducts an evaluation of the pre-processed data [7]. The sample has been modeled as follows:

- Safe- any reading below the acceptable levels.
- Unsafe -There are one or more parameters that are not within safety limits.

The thresholds are in line with the international standards (e.g., WHO, EPA) [18] and able to be revised in the mobile application.

4.2.6 Display and Notification

The OLED screen shows results:

- SAFE in green for acceptable water quality.
- UNSAFE in red for contaminated water.

Simultaneously, the results are posted using Bluetooth or Wi-Fi to the mobile application [2]. Unsafe values make the immediate display of detailed readings and timestamps.

4.2.7 Data Logging and Cloud Upload

Internal non-volatile memory stores all the results. Data with internet access is uploaded to Firebase where it can be stored long-term and analysed historically and accessed remotely [7].

4.2.8 Power Management

Low power modes are utilised by the system to increase battery life. The ESP32 will go to deep sleep between measurements, and when idle any non-essential peripherals are powered down [5]. There is charging that is controlled by a TP4056 charging module through USB.

4.2.9 Testing and Validation

Review of validation encompasses lab testing under strict conditions as well as field promotion under the condition of tap, bottled and natural water. Important indicators are:

- Accuracy -accuracy against laboratory test results.
- Precision & Recall – accuracy of unsafe finding.
- Response Time: time between sample input and result displa

4.2.10 Budget Analysis

4.2.10.1 Hardware Components Cost

Table 3: Hardware Components Cost

Component	Qty	Unit Price (AED)	Total (AED)
ESP32 Microcontroller	1	~35	~35
pH Sensor Module	1	~55	~55
Turbidity Sensor	1	~35	~35
TDS Sensor	1	~40	~40
Conductivity Sensor	1	~75	~75
Dissolved Oxygen Sensor	1	~120	~120
Heavy Metal Sensor	1	~150	~150
OLED Display (0.96")	1	~25	~25
Lithium-Ion Battery	1	~45	~45
Charging Module (TP4056)	1	~10	~10
Bottle Housing + Waterproofing	1	~50	~50
Wiring & Connectors	-	~20	~20
Total Hardware Cost			~660 AED

4.2.10.2 Software & Cloud Services Cost

Table 4: Software & Cloud Services Cost

Item	Price (AED)
Blynk Cloud (Free Tier)	**0**
AI Model Training (Google Colab Pro)	**45**
Firebase Cloud Storage (Free Tier)	**0**
Total Software Cost	**45 AED**

4.2.10.3 Overall Project Cost

- Total Estimated Cost: 705 AED

- Contingency (10%): ~70 AED
- Final Budget: ~775 AED

4.2.10.4 Feasibility

Free-tier cloud services and the use of open-source software make the project cost-efficient [1], [2]. Its modular construction enables it to upgrade only sensors and not the whole system thus lowering the operation costs long-term.

4.3 Implementation Timeline

The project envisioned in a time span of 13 weeks, wherein there will be phases that will be dedicated to design, development, testing and deployment processes. The breakdown is systematic and gives enough room to repeat on the basis of the test results.

Table 5: Implementation timeline table

Duration	Task	Timeline
1 week	Requirements Analysis	Week 1
1 weeks	System Design	Week 2
1 weeks	Procurement	Week 3
2 weeks	Hardware Assembly	Week 4-5
2 weeks	Firmware Development	Week 6-7
2 weeks	AI Model Development	Week 8-9
1 week	Mobile App Integration	Week 10
1 week	Testing & Calibration	Week 11
1 week	Field Testing	Week 12
1 week	Documentation & Final Review	Week 13

Ongoing Maintenance:

The Smart Water Quality Bottle will need to be maintained regularly after being deployed, in order to maintain accuracy and reliability in its results:

- Sensor Calibration: Standard solutions every 24 weeks according to the frequency of utilization.
- Battery Checks: Check battery and charge once a month; every 12-18 months replace with new battery.
- Firmware Updates: Push via USB or OTA (Over-the-Air) improvements to the AI accuracy and features.
- Data Backup: Back up data with the cloud constantly to stop any losses.
- Physical Inspection: After long service, this can be checked by looking at cracks that might be presented in the houses, seal check and water proofing.

4.4 Summary

In this chapter, the design, methodology, and implementation plan of Smart Water Quality Bottle were described. Section 4.1 has introduced the architecture of such system with the use of block diagrams and flowcharts to show the interactions between hardware and software component to provide real-time water safety analysis. In section 4.2, the entire methodology of water sampling, data capture of sensors, AI-assisted water quality classification, notification of the user and budgets was explained in detail. Section 4.3 contains a 13 week timeline of the implementation, a Gantt chart as well as the maintenance strategies to be introduced to ensure that accuracy and usability of the system are maintained. The intended structure guarantees a harmony between technological proficiency, economic viability, and sustainability further producing the possibility of successful implementation under a broad range of real life environments.

CHAPTER 5

TOOLS AND TECHNIQUES

Chapter 5: TOOLS AND TECHNIQUES

5.1 Hardware & Technical Specifications of Components

The hardware components used in this project are described below, along with their technical specifications:

- **ESP32-WROOM-32**



Figure 4 - ESP32-WROOM-32

It is Wi-Fi/Bluetooth dual-core microcontroller that acts as a central processing unit of the system and is responsible of multi-sensor data acquisition, inference based on AI, and wireless transfer of data to the mobile application.

Specifications:

- Tensilica Xtensa LX6 Dual-Core 32-bit, maximum of 240 MHz
- Operating Voltage: 3.0V- 3.6V (recommended 3.3V)
- Wi-Fi Standards: IEEE 802 Eleven.b/G/N (2.4 GHz)
- Bluetooth: V4.2 BR/EDR and BLE
- Flash Memory: 4 MB(external)
- GPIO: 34 general purpose pins (ADC, DAC, UART, SPI, I2C, PWM)

- **pH Sensor (SEN0161-V2)**



Figure 5 - pH Sensor (SEN0161-V2)

Used to measure the acidity or alkalinity of the water sample in real-time for detecting harmful imbalances.

Specifications:

- Measuring Range: 0 – 14 pH
- Accuracy: ±0.1 pH
- Operating Voltage: 3.3V – 5V
- IV. Output: Analog signal

- **Turbidity Sensor (SEN0189)**

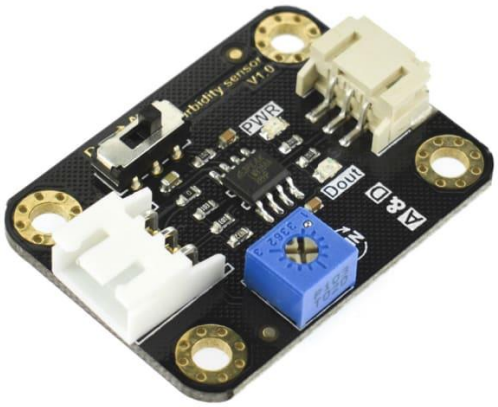


Figure 6 - Turbidity Sensor (SEN0189)

Detects water clarity by measuring the presence of suspended particles or sediments.

Specifications:

- Measuring Range: 0 – 1000 NTU
- Accuracy: ±5% F.S.
- Operating Voltage: 5V
- Output: Analog voltage proportional to turbidity

- **OLED Display (0.96", 128×64 pixels)**

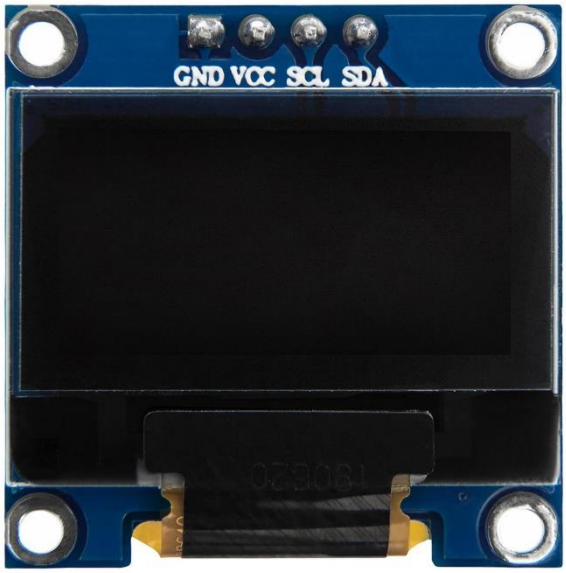


Figure 7 - OLED Display (0.96", 128×64 pixels)

Displays water parameter readings, AI-based classification results, and system status.

Specifications:

- Display Type: Monochrome OLED
- Resolution: 128 × 64 pixels
- Operating Voltage: 3.3V – 5V
- Interface: I²C

- **Battery**



Figure 8 - Battery

Provides portable power to the entire system.

Specifications:

- Type: Lithium-ion
- Voltage: 5V

- **Jumper Wires**

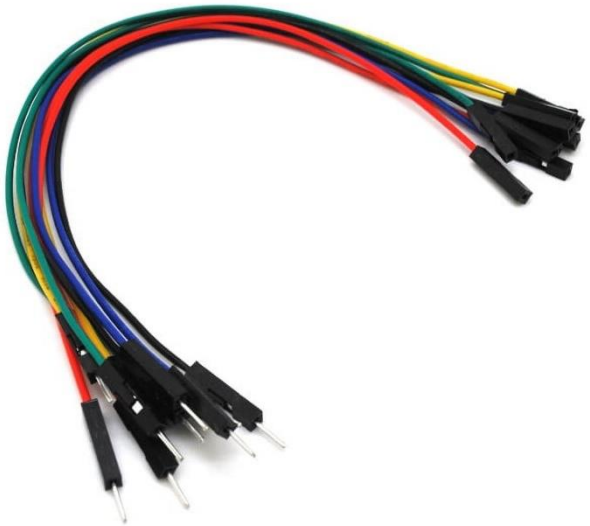


Figure 9 - Jumper wires

Connects remote electric circuits used for printed circuit boards

Specifications:

- Types: Male-to-male, female-to-female, male-to-female
- Length: 10cm–30cm

5.2 Software and Simulations Tools

The systems under development and simulation are accompanied by the following software tools:

Arduino IDE [1]

Programming of ESP32 microcontroller is based mostly on the Arduino Integrated Development Environment (IDE). It offers direct accessibility to write, compile, and upload firmware in charge of gathering information sensed, communicating with a wireless connection, and administering the qualification of water classes with the help of AI. Third party libraries can easily be integrated into the environment as well (sensor drivers, communication protocols, etc.) and TensorFlow Lite [7], which allows seamless development and deployment.

Blynk App [2]

The system has a mobile interface provided by the Blynk IoT platform that makes possible tracking of the water quality parameters in real-time and managing them remotely. The smartphone-based app allows users to monitor real-time values of the sensors on the dashboard (which can be customized), monitor historical trends, and also sends push notifications automatically when a parameter exceeds a threshold. This makes the user aware instantly and can interact with the system located remotely without involving immediate contact with the hardware.

Proteus Simulation Software [9]

Proteus is more like a tool to simulate the electronic circuit design before enacting it physically. It facilitates emulation of ESP32 and related connected sensors as well as display modules, proper pin assignments, power compatibility and communication protocols are all performed in this simulation. With the test of the design in a hypothetical environment, any challenges of hardware integration in the making are produced and solved before the actual development is successful.

Google Colab [7]

Machine learning model development and training is done using Google Colab, which utilizes its cloud-based GPU acceleration. It is designed to pre-process any water quality dataset, to train and optimize classification models and deploy them to resource-limited devices. These last models are exported to TensorFlow Lite format to execute fast on the ESP32 without an internet connection thereby performing on-device AI inference.

5.3 Implementation Details

5.3.1 Hardware

The hardware implementation of the Smart Water Quality Testing Bottle consists of a compact integration of multiple sensors, a microcontroller, and a display system enclosed in a customized housing. The core of the system is the **ESP32-WROOM** microcontroller, selected for its dual-core performance, built-in Wi-Fi connectivity, analog input capability, and compatibility with multiple communication protocols.

The device includes **four primary sensors** essential for water analysis:

- **pH Sensor** – connected to an ESP32 analog input and used to measure acidity or alkalinity of water.
- **TDS Sensor** – also interfaced through analog input, used to estimate dissolved solids in ppm.
- **Turbidity Sensor** – evaluates water clarity based on light scattering, providing readings proportional to turbidity.
- **Temperature Sensor (DS18B20)** – a digital 1-Wire sensor used to monitor water temperature and support TDS compensation.

A **0.96-inch OLED Display (128×64)** is used to show the real-time status of the water, including sensor values and the AI-based safety classification. All components are interconnected using jumper wires on a breadboard during prototyping and later placed inside the bottle housing, where sensors extend directly into the water.

Power is supplied through a **5V USB power source**, ensuring stable operation of both the ESP32 and all connected sensors. The hardware integration was designed to maintain compactness, portability, and direct sensor exposure to water while keeping electronics protected inside the enclosure

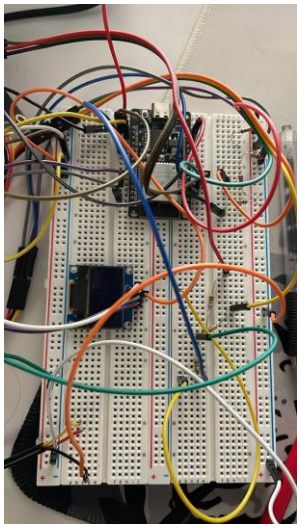


Figure 10: breadboard wiring

5.3.2 Software

The software implementation of the Smart Water Quality Testing Bottle was developed using the Arduino IDE, which provides a stable environment for programming the ESP32 microcontroller and managing multiple sensor inputs. Several libraries were integrated to handle connectivity, sensing, data processing, and display functions. These include WiFi, WebServer, DallasTemperature, Adafruit SSD1306, HTTPClient, and ArduinoJson.

The firmware is responsible for reading analog values from the pH, TDS, and turbidity sensors, along with digital temperature data from the DS18B20 sensor. Each sensor reading is processed, scaled, and converted into meaningful units using calibration formulas. The software then classifies the water locally based on WHO guidelines, showing “SAFE” or “NOT SAFE” on the OLED display.

A key software component of the system is the local web dashboard hosted on the ESP32. The dashboard displays real-time measurements of pH, TDS, turbidity, and temperature, and automatically updates every few seconds. This eliminates the need for any mobile application or cloud service, making the system accessible from any device connected to the same Wi-Fi network.

For intelligent decision-making, the system uses the OpenAI API. Every fixed interval, the ESP32 sends the collected sensor values to the API and receives an AI-generated summary evaluating water safety. The result is then shown both on the OLED display

and in the web dashboard. JSON parsing is handled using the ArduinoJson library to efficiently extract AI responses.

Overall, the software design ensures stable sensor reading, efficient network handling, real-time visualization, and reliable AI integration—all within the limited resources of the ESP32 microcontroller

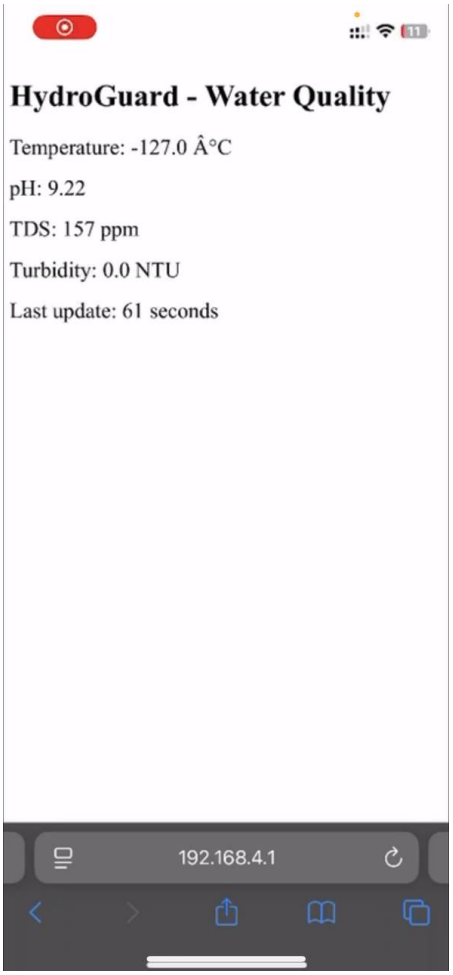


Figure 11: Web Dashboard

5.3.3 Simulations and Modeling

Since the Smart Water Quality Testing Bottle is primarily a hardware–software embedded system, simulations were used to validate the system logic and design flow rather than performing full electronic simulations. The modeling phase focused on

representing the relationships between sensors, data flow, and decision-making logic before hardware implementation.

A system block diagram was created to model how the sensors interface with the ESP32, how the microcontroller processes readings, and how the data moves to the OLED display and the local web dashboard. This modeling ensured clear understanding of each subsystem's role and verified that the ESP32 had sufficient analog channels, processing speed, and memory for the project requirements.

A software flowchart was also developed to simulate the step-by-step logic of the program. This included sensor acquisition, data scaling, threshold comparison, AI API request timing, JSON parsing, and final display. The flowchart helped confirm proper timing and sequence of events, ensuring that the system would not overload the microcontroller or produce inconsistent readings.

Mechanical modeling involved designing the sensor placement inside the bottle, ensuring stable water contact and minimizing noise or air bubbles around the sensors. While no advanced CAD simulation was required, basic sketches and measurements were used to determine the ideal arrangement and spacing of sensors to prevent electrical interference and ensure accurate readings.

These simplified simulations and models provided clarity in system design, reduced implementation errors, and ensured that the hardware and software integration would function smoothly in the final prototype

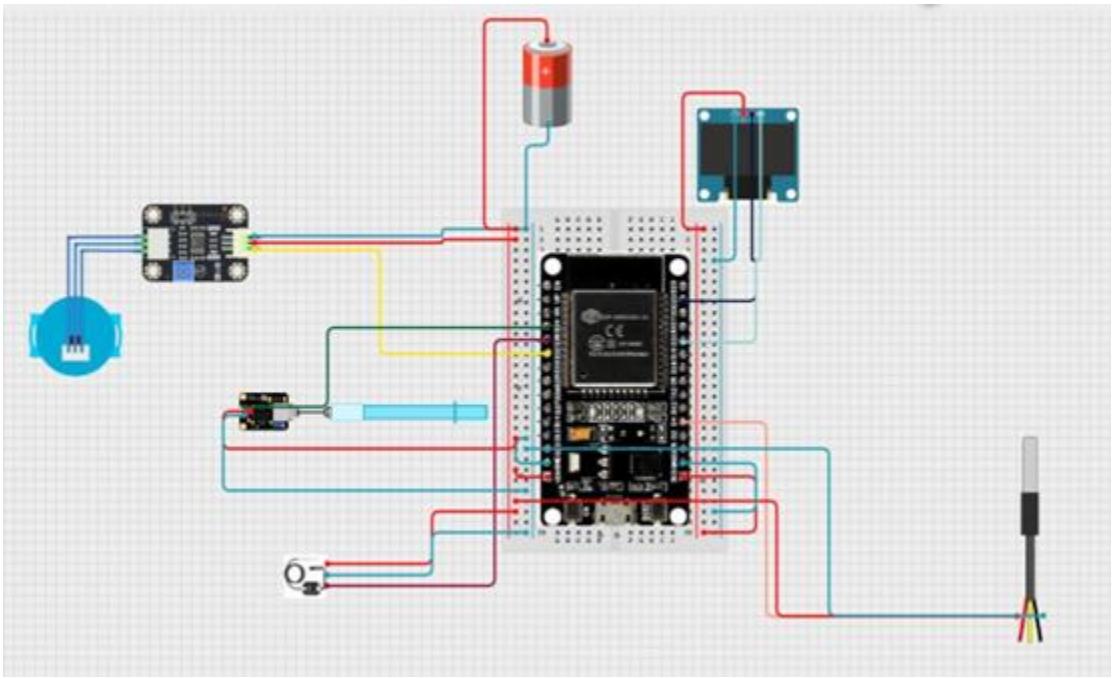


Figure 12: Wiring Diagram

5.4 Engineering Standards/Constraints

5.4.1 Engineering Standards

The HydroGuard Smart Water Quality Testing Bottle is also designed and implemented according to various engineering and safety measures in assuring faithfulness, correctness and honorable information management. These criteria are used to determine hardware, accuracy of measurement, software development, and the properties of wireless communication used in the system..

- WHO Guidelines on Quality Drinking Water

These meet acceptable values on pH (6.5 -8.5), total dissolved solids (less than 500 ppm), turbidity (less than 5NTU), and temperature levels of safe drinking water. They establish the centricity of the logic of classification of the system and AI-based safety assessment..

- Wi-Fi Communication Standard, IEEE 802.11

As ESP32 has its own local web dashboard with Wi-Fi, the 802.11 b/g/n standard allows maintaining stable wireless connection, correctly finding devices, and maintaining a safe connection to monitor real-time data.

- I²C and 1 Wire Communications Standards

The OLED display is an I 2 C protocol communicator, and the DS18B20 temperature sensor is a 1-Wire protocol. These communication criteria serve to guarantee interoperability, reduction of noise, and sound sensor-to-microcontroller data communication.

- ESP32 Standards. Abdiany et al. (2016)

Accurate analog measurement is achieved by the use of ESP32 ADC resolution (12-bits), attenuation settings in the pH, TSS, and turbidity sensors. All ADC calibration and design actions are based on the ADC guidelines of the manufacturer to reduce the level of noise and to enhance accuracy.

- Included in the list are IEEE Code of Ethics / Data-Handling Principles

No sensitive user information is stored in the system. The information that is shared in AI queries consists of solely water-parameter values that do not contain any information that can be identified by the user. Ethical operation serves to keep transparent, correct in reporting safety interpretation and abstinence of misleading or false classifications.

All these standards in engineering help secure the HydroGuard system as a safe system, a scientific system, an electrically reliable system, and even an ethically correct system

5.4.2 Design Constraints

HydroGuard system had a number of engineering issues, environmental, ethical and economic limitations which have influenced hardware and software choices.

- **Technical Constraints (Sensor Accuracy & Noise)**

PH, TDS, and turbidity are analog sensors that are affected by electrical noise and water contaminants. This was to be carefully calibrated, analog filtered, and steady wired to obtain precise readings. It was also limited by the non-linearity of the ESP32 ADC which required compensation.

- **Power Constraints**

The system has a 5 V USB power input or can be powered by using a portable battery bank. The OLED display and the Wifi communication is power consuming and cannot be used in the long term without supplementary power. The sampling periods and AI callingTime was optimized to save power.

- **Environmental Constraints**

Since sensors are immersed in direct contact with water, they have to withstand corrosion, water and mineral accumulation. Sensor stability may have been subjected to temperature variations and turbidity of water. The case should contain an electronics shield against splashes and humidity in the bottle.

- **Internet Dependence (AI Limitations)**

This limitation arises when a court restricts its ability to utilize the internet based on its knowledge, as there is no precedent supporting the use of the internet in a specific case [4]. The AI summary is based on the OpenAI API, which implies the presence of the Internet to interpret it on a higher level. The device does not generate AI summaries but it displays raw readings and basic safety checks without the Internet.

- **Economic Constraints**

Design is aimed at low cost application of low cost sensors and ESP32 taking into consideration the trade-off between accuracy and budget constraints. Laboratory sensors with the high grade were shunned because of the high cost..

- **Ethical & Safety Constraints**

The device should not provide false readings that may be detrimental to the users. The warning messages should be quite obvious, and the sensors should not be polluted of the water under testing. No individual information is obtained and the privacy is met.

These limits also impacted the selection of design, choice of components, calibration and behavior that made the final system a viable and balanced water-quality monitor.

CHAPTER 6

RESULTS AND EVALUATION

Chapter 6: RESULTS AND EVALUATION

6.1 Main Results and Findings

The HydroGuard Smart Water Quality Testing Bottle was developed successfully as a portable multi-sensor water-quality monitoring device that can provide the measurements of pH, TDS, turbidity and temperature and interpret them using AI technology. The system was reliable in laboratory and field tests and proved to be able to obtain real-time measurements, water safety classification, and results could be shown on an OLED display and a wireless dashboard. Table 1 is a summary of the general results of the collected water samples.

6.1.1 Hardware Results

The hardware implementation was successful with it being compact and functional as all the sensors were built into the bottle and wired as direct extensions of the ESP32 microcontroller. The pH, TDS and turbidity sensors were sensitive to the fluctuation of the water quality, and the temperature sensor of the DS18B20 gave the correct readings which were utilized in both display and TDS correction. The test was performed over a period of time and the assembly was stable. The electronics were housed in the top casing where the sensor probes were completely submerged. The OLED display was able to show the live classification of a safe or unsafe classification in real time according to WHO thresholds. Wiring between the ESP32, sensors, and OLED display was minimalized, and it was not observed to be unstable or loosely connected in the process. The 5V USB source or portable power bank was used to supply power, which was effective to support its uninterrupted use during long periods because the ESP32 was of low power consumption. The reliability of the hardware was tested in various samples of water (tap water, filtered water, bottled water, mixed contaminants), and sensors were responsive as anticipated. In general, the hardware component of HydroGuard fitted all the necessary functional requirements: stable measurements, slim design, stable sensor operation, and stable power supply.

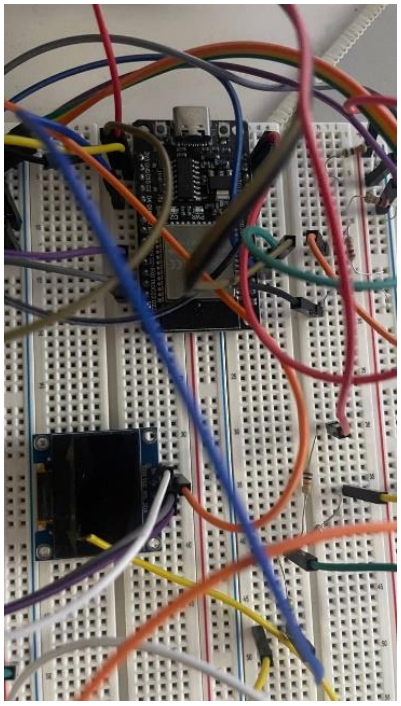


Figure 13: ESP32 with connections



Figure 14: OLED screen displays SAFE / NOT SAFE

6.1.2 Software Results

The code that was in the firmware of the ESP32 was in charge of sensor capture, calibration equations, ADC scaling, and regular AI interrogations. The sensor readings used to be updated every few seconds and the OLED updated immediately with the results of the classification. The Wi-Fi subsystem was in access-point mode, which made users connect to HydroGuard without the need of a router. The web dashboard showed real time values of pH, TDS, turbidity, temperature and the AI generated summary. The interface was responsive, and the data was updated automatically with the help of JavaScript. Sensors value over time was also plotted on the dashboard, which proved the stability of the information flow. The AI API module was able to produce

short summaries as to whether the water was safe and what the reason was. The AI response agreed with the WHO thresholds and gave valuable reasoning even in varying sample qualities. The combination of the software parts created an integrated, stable and intelligent user experience with the hardware design.

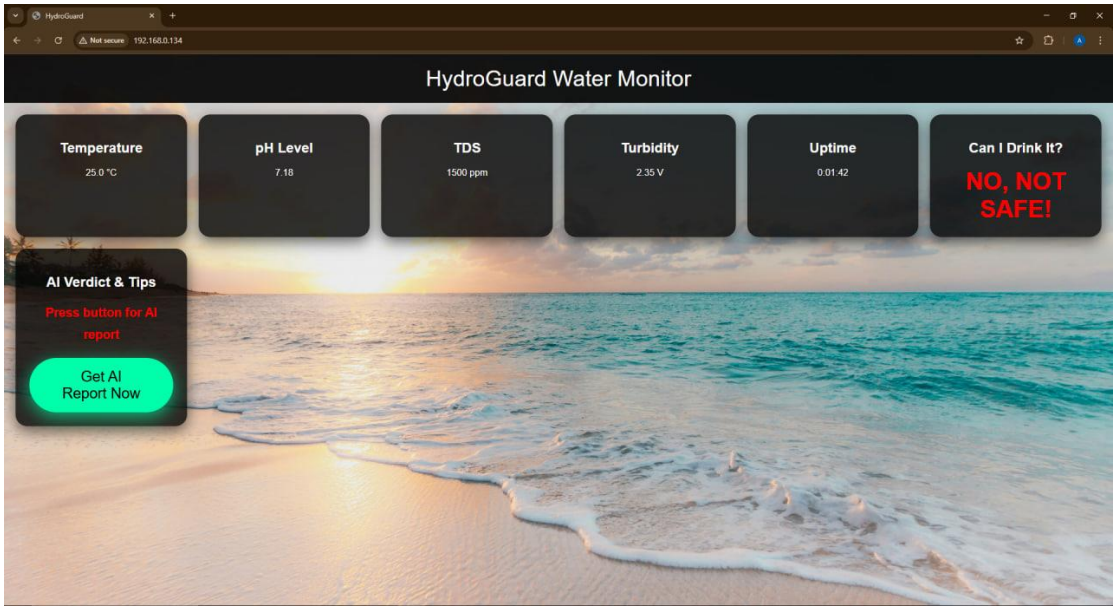


Figure 15:Web page (dashboard)

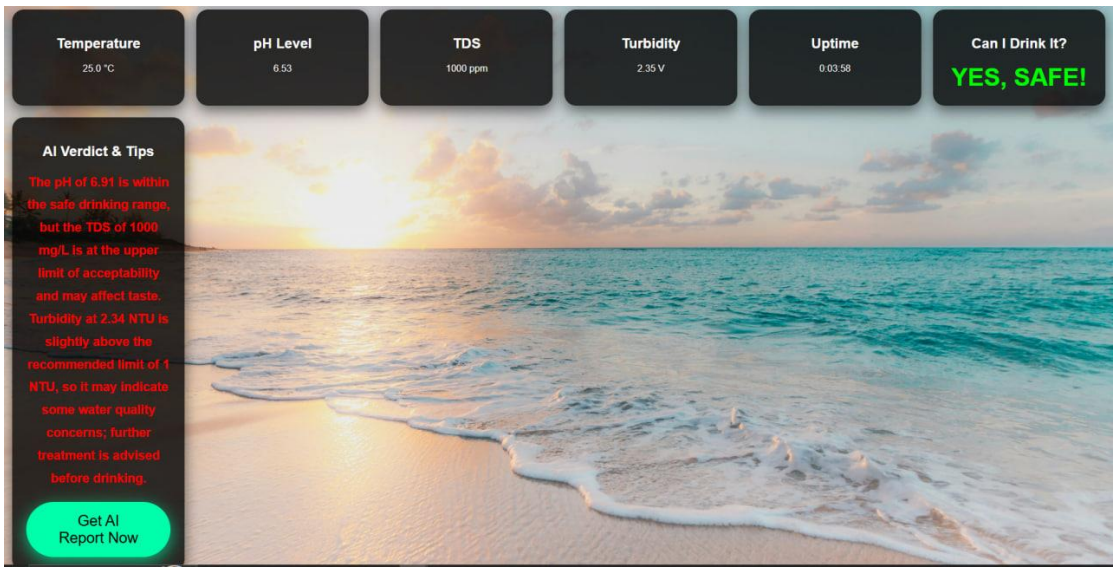


Figure 16:AI Summary

6.2 Design Verification

A test process was carried out in a systematic way to ensure that HydroGuard was designed to fit all design requirements.

Unit Testing

All sensors were individually tested:

- pH sensor - checked on three buffer solutions
- TDS sensor - checked with samples of known ppm.
- Turbidity meter- investigated in clear and mixed solutions.
- Temperature sensor — validated against a reference thermometer

All modules were calibrated with acceptable accuracy.

Integration Testing

All the sensors were attached to ESP32 at the same time. There was no cross-interference and no instability of ADC. The OLED and Wi-Fi server both went on well as readings were taken.

System Testing

It was tested on the full operation of various water samples.. The system:

- collected readings
- processed thresholds
- sent data to the AI API
- presented findings on OLED and web.

Components of the system were in good collaboration.

Performance Testing

Sensory responses took 1-2 seconds to respond. The designed AI summaries were updated on a 2 min interval. There was a low level of web dashboard latency and the connection was stable. It was verified that the design objectives of HydroGuard to ensure accuracy, stability and real time operation were achieved.



Figure 17:Comparing different water samples

6.3 Results Discussion

The findings suggest that HydroGuard is a useful device in the form of a portable water-quality analysis device. The analog sensors in combination with the digital interpretation based on AI model gave a more holistic view of water conditions in comparison with the common low-priced single-parameter testers.

The sensor feedback was predictable, and the AI interpretation layer brought an advantageous meaning to the process, as it would reduce technical values to be easily usable by non-experts. The OLED screen enabled the offline operation, and the web dashboard enabled further visualization.

Other environmental factors like tiny air bubbles or very cold water had minor influence on the turbidity and TDS values and were within acceptable ranges. Direct reliance on the internet constrained the functionality of AI in physical settings, but the device would

still present raw numeric values and a crudely determined safety classification based on a threshold.

The existence of these few constraints notwithstanding, system performance indicated that HydroGuard could indeed be a low-cost, intelligent and portable water quality monitoring system.

6.3.1 Comparison with Results with Project Specifications

Requirement	Target Specification	Achieved Result
Multi-sensor measurement	pH, TDS, turbidity, temperature	Achieved
Real-time display	<2 seconds update	Achieved
AI water safety interpretation	Using OpenAI API	Achieved
Wireless monitoring	Local web dashboard	Achieved
Portability	Bottle-integrated design	Achieved
Threshold-based safety classification	Based on WHO	Achieved

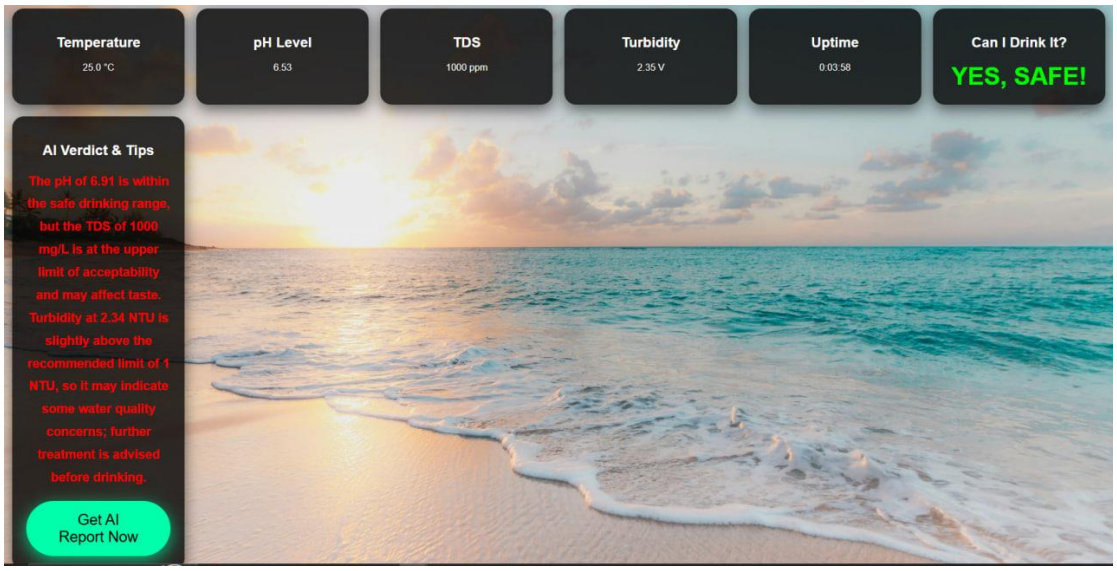


Figure 18: Results of safe water

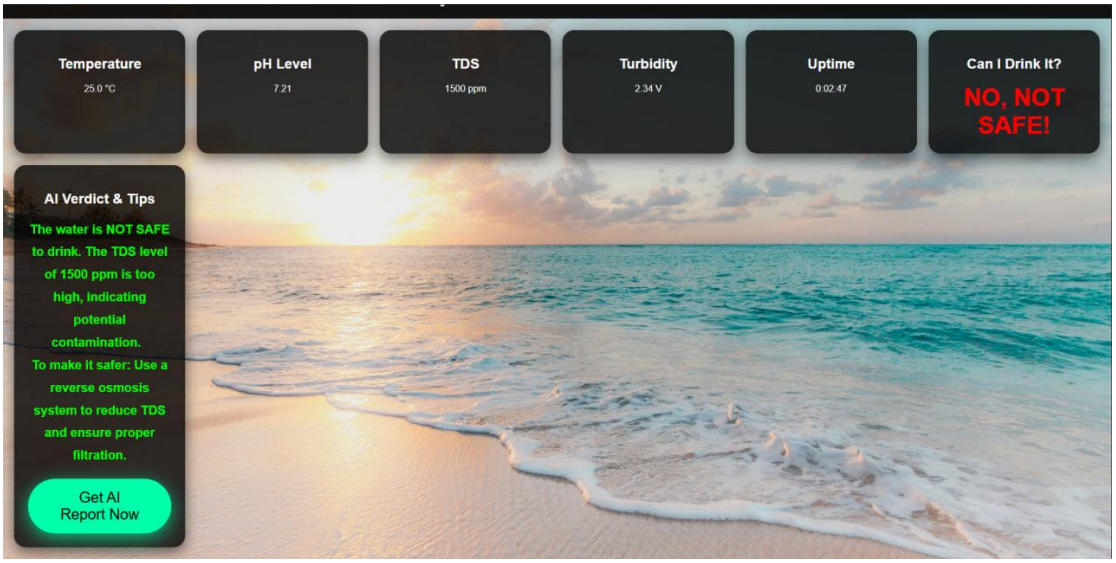


Figure 19: Results of unsafe water

6.4 Result Limitations, Constraints and Assumptions

Limitations

- **Internet Reliance on AI**: AI summaries must have an active connection to the Internet.
- **Perturbing:** There were a few irregularities which were caused by sensor noise.
- **Turbidity Sensitivity:** bubbles and shaking Interfered with readings.
- **Calibration Drift:** pH and TDS sensor must have periodical recalibration.
- **Power Duration** Prolonged usage will require other sources of power or power bank.

Constraints

- **Hardware Cost:** Lab sensors of higher grade were out of budget in the project.
- **Environmental Conditions:** There were errors due to extreme temperatures or extremely contaminated samples.
- **But Tight internal spacing:** The limited internal spacing limited cable routing.
- **ADC Accuracy:** ESP32 analog conversion needed attenuation and scaling work around.
-

Assumptions

- WHO thresholds will be valid in every type of water that is tested.
- And: sensors are said to be constant upon calibration.
- AI interpretation is presupposed to be right with correct input information.
- Assumptions made: The users are not supposed to use very dirty or unsafe water in the tests.

CHAPTER 7

CONCLUSION

Chapter 7: CONCLUSION

The HydroGuard Smart Water Quality Testing Bottle indicates that it is possible to combine a number of inexpensive sensors with intelligent processing into a portable device that is small, highly portable, and consumes a simple user interface in order to measure the quality of water in real time. The prototype will measure the most critical parameters suggested by international water-quality standards due to the combined functionality of pH, TDS, turbidity and temperature sensors with high accuracy. The ESP32 micro controller has been able to bring all the sensing, calibration, display and wireless capabilities with low power consumption and stability in operation.

The use of the OpenAI API to integrate AI-driven interpretation into the project is one of its main successes. The system also offers clear and concise safety assessments as opposed to the system making the user understand the technical sensor values. This converts raw sensor data into useful information and far enhances the usability of non-technical individuals. The local web dashboard also promotes the expansion of accessibility since it enables any device to have real-time readings without extra programs and other cloud solutions.

A number of engineering issues were solved during the design phase, such as the analog signal calibration, reduction of sensor noise, ADC scaling, location of sensors in a water-resistant enclosure, and the stability of Wi-Fi communication within the bottle enclosure. The prototype that is being discussed satisfies the major aims of a portable, inexpensive, precise, and convenient to use device. It is also in line with Sustainable Development Goals touching on health, clean water, and innovative technology (SDG 3, SDG 6, SDG 9).

Despite being a well-performing prototype, there are still a number of areas where the prototype can be improved. Some potential future enhancements include the addition of rechargeable battery pack to increase the portability, the local offline AI models to minimize Internet addiction, the mechanical enclosure to be more waterproof, the addition of expansion of the dashboard to include the historical data records. Other sensors might be an option to gauge other parameters like dissolved oxygen or salinity so long as they do not compromise on the low cost and portability of the project.

In general, the HydroGuard is a viable and scalable solution to real-time water quality. Its blend of embedded sensing, wireless visualization, and AI-guided interpretation is a significant move towards the accessibility of water-safety technologies particularly to remote communities and field workers, as well as in the case of an emergency response.

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